

GOVERNANCE HEALTH HEATMAP – FORENSIC EVIDENCE DOSSIER

COURT-ADMISSIBLE ELECTRONIC RECORD • ETDA B.E. 2544 SECTION 28 • NIST FIPS 204 ML-DSA-87

LIVE ROOT [CH-02 WORM]: 9088d5b5fb073482e73ff2a43f0f19c127511f261d4ad87... | BLOCK #849202  
P R I N C I P A L : 0<ÿü0<ÿí 2 " " 8 9 ! 4 2 @ 5 " # ( E v i d e n c e L e d g e r S e a l e r ) ( #

SECTION 1: ETDA B.E. 2544 SECTION 28 EVIDENTIARY CHAIN OF CUSTODY

Dossier ID: ETDA-SEC28-HEATMAP-1790578347277 | Captured UTC: 2026-09-28T06:52:27.277Z

Grid Mode: SEAL\_STATUS | Active Metric: COHERENCE | Cycle: #849202

HSM Quorum State: 7/10 REAL\_HSM (HIGH-PRIORITY BREACH ALERT <8/10)

SHA-256 State Digest: 0x3fb7339c8cfd16c4909ab814479844d8a14816bed34cdbb07528e18501da86fc

HIGH-PRIORITY HEALTH BREACH ALERT: ISOLATED HSM NODES (7/10 ACTIVE < 8 THRESHOLD)

| Node ID | Hardware Signer                    | Dossier Reference       | Chamber                          | Public Key                 | Status          |
|---------|------------------------------------|-------------------------|----------------------------------|----------------------------|-----------------|
| TC-03   | HSM Node #03 (Sub-Kelvin Sentinel) | DOSSIER-HSM-TC03-849202 | CH-04 (Merkle Attestation)       | dilithium5_pk_33f990a14b88 | ISOLATED_BREACH |
| TC-08   | HSM Node #02 (TC-08)               | DOSSIER-HSM-TC08-849202 | CH-05 (Deca-Key Quorum Vault)    | sphincs_pk_cc274199a012    | ISOLATED_BREACH |
| TC-09   | HSM Node #03 (TC-09)               | DOSSIER-HSM-TC09-849202 | CH-15 (Cold Quorum 72h Vault)    | dilithium5_pk_e2d49500b182 | ISOLATED_BREACH |
| TC-10   | HSM Node #01 (TC-10)               | DOSSIER-HSM-TC10-849202 | CH-17 (Executive Synthesis Core) | falcon1024_pk_7789f811cb90 | ISOLATED_BREACH |

SECTION 2: 18 SOVEREIGN CHAMBERS – HARDWARE SEAL STATUS & INTEGRATION COVERAGE MATRIX

| Chamber | Sovereign Designation    | Seal Status | Coherence | Cryo Temp | E2E Stage | Lines/Br Cov  | E2E Pass | Uncovered Path Lines                                  |
|---------|--------------------------|-------------|-----------|-----------|-----------|---------------|----------|-------------------------------------------------------|
| CH-00   | Sovereign Foundation & G | PURE_GREEN  | 99.99%    | 14.98 mK  | REQUEST   | 100.0% / 92%  | 8/8      | None (Nominal & Disconnected Provider paths verified) |
| CH-01   | Multi-Key Cryptographic  | PURE_GREEN  | 100.00%   | 15.05 mK  | PROPOSAL  | 96.4% / 83%   | 8/8      | 1012-1015                                             |
| CH-02   | Immutable Audit Ledger & | PURE_GREEN  | 99.99%    | 15.05 mK  | PROPOSAL  | 93.6% / 73%   | 8/8      | L1309-1324                                            |
| CH-03   | Safe Harbor Compliance & | PURE_GREEN  | 99.99%    | 14.99 mK  | EXECUTE   | 95.1% / 88%   | 8/8      | 918-924                                               |
| CH-04   | Real-Time HSM Quorum (De | PURE_GREEN  | 99.99%    | 14.92 mK  | VERIFY    | 40.0% / 50%   | 6/8      | 198-312                                               |
| CH-05   | 6-Stage DAG Execution En | PURE_GREEN  | 99.99%    | 14.90 mK  | APPROVAL  | 36.8% / 50%   | 6/8      | 1539-1576                                             |
| CH-06   | Circuit Breaker & Fail-C | PURE_GREEN  | 100.00%   | 14.96 mK  | ANALYSIS  | 100.0% / 100% | 8/8      | None (ZYRQUEN_CORE write block verified)              |
| CH-07   | Quantum Continuum & Phoe | PURE_GREEN  | 99.99%    | 15.03 mK  | AUDIT     | 98.6% / 64%   | 8/8      | None (100% E2E Verified)                              |
| CH-08   | Merkle Tree SSoT Verifie | PURE_GREEN  | 99.99%    | 15.06 mK  | ANALYSIS  | 65.8% / 45%   | 6/8      | 1305-1341                                             |
| CH-09   | Defense-Grade High Assur | PURE_GREEN  | 100.00%   | 15.01 mK  | ANALYSIS  | 92.5% / 74%   | 8/8      | L1323-1338                                            |
| CH-10   | Sovereign Treasury & Bud | PURE_GREEN  | 99.99%    | 14.94 mK  | TARGET    | 21.8% / 2%    | 6/8      | 836, 848, 861-894                                     |
| CH-11   | Court-Admissible Dossier | PURE_GREEN  | 99.99%    | 14.90 mK  | APPROVAL  | 86.4% / 85%   | 7/8      | L1327-1342                                            |
| CH-12   | Zero-Trust Write Firewal | PURE_GREEN  | 99.99%    | 14.94 mK  | EXECUTE   | 90.1% / 90%   | 8/8      | L1329-1344                                            |

| Chamber | Sovereign Designation    | Seal Status | Coherence | Cryo Temp | E2E Stage | Lines/Br Cov | E2E Pass | Uncovered Path Lines                                  |
|---------|--------------------------|-------------|-----------|-----------|-----------|--------------|----------|-------------------------------------------------------|
| CH-13   | Distributed Quorum Conse | PURE_GREEN  | 99.99%    | 15.01 mK  | TARGET    | 93.8% / 95%  | 8/8      | L1331-1346                                            |
| CH-14   | Neural & Heuristic Anoma | PURE_GREEN  | 100.00%   | 15.06 mK  | VERIFY    | 97.5% / 66%  | 8/8      | None (100% E2E Verified)                              |
| CH-15   | Sonic Alert & Multilingu | PURE_GREEN  | 99.99%    | 15.03 mK  | AUDIT     | 98.5% / 90%  | 8/8      | 1338-1340                                             |
| CH-16   | Dynamic 3D Sovereign Qua | PURE_GREEN  | 99.99%    | 14.96 mK  | REQUEST   | 91.4% / 76%  | 8/8      | L1337-1352                                            |
| CH-17   | Supreme Omnipresent Comm | PURE_GREEN  | 100.00%   | 14.90 mK  | ANALYSIS  | 95.1% / 81%  | 8/8      | None (100% E2E Verified)                              |
| CH-18   | Neural Sentinel & Predic | PURE_GREEN  | 99.99%    | 14.92 mK  | REQUEST   | 100.0% / 92% | 8/8      | None (Nominal & Disconnected Provider paths verified) |

**SECTION 3: PRODUCTION INTEGRATION COVERAGE & 14,902 SEALS ATTESTATION**

V8 Coverage Summary: Statements 85.46% | Branches 60.15% | Functions 95.83% | Lines 86.22%

Canonical Hardware Seals: 14,902 SSoT | Compliance: 99.999% | Untested Path Cells: 15

Legal Declaration: Certified authentic electronic record pursuant to Thailand ETDA B.E. 2544 Sections 9, 26, 28.